MiniNumPy Library

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Chapter 1

Namespace Index

1.1 Package List

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Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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A simplified multi-dimensional array class, similar to NumPy's ndarray 17

Chapter 3

File Index

3.1 File List

Here is a list of all files with brief descriptions:

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Chapter 4

Namespace Documentation

4.1 mininumpy Namespace Reference

MiniNumPy package initializer.

Namespaces

- namespace array
- namespace linalg

  MiniNumPy Linear Algebra Module.

4.1.1 Detailed Description

MiniNumPy package initializer.

This file makes the mininumpy directory a Python package and exposes key classes and functions for easier access.

4.2 mininumpy.array Namespace Reference

Classes

- class Array

  A simplified multi-dimensional array class, similar to NumPy's ndarray.

Functions

- array (data)

  Creates an Array object from a list or nested list.
- _create_nested_list (shape, value)

  Helper to create a nested list of a given shape filled with a value.
- zeros (shape)

  Creates an array of a given shape, filled with zeros.
- ones (shape)

  Creates an array of a given shape, filled with ones.
- eye (n)

  Creates an N x N identity matrix.
- arange (start, stop=None, step=1)

  Creates an array with a range of values.
- linspace (start, stop, num)

  Creates an array with a specified number of evenly spaced values.
- _apply_func_to_array (a, func)

  Helper to apply a function elementwise to an array.
- exp (a)

  Computes the exponential of all elements in the input array.
- log (a)

  Computes the natural logarithm of all elements in the input array.
- sqrt (a)

  Computes the non-negative square-root of all elements.
- abs (a)

  Computes the absolute value of all elements.
- sum (a)

  Computes the sum of all array elements.
- mean (a)

  Computes the arithmetic mean of all array elements.
- min (a)

  Finds the minimum value among all array elements.
- max (a)

  Finds the maximum value among all array elements.
- argmin (a)

  Finds the index of the minimum value in the flattened array.
- argmax (a)

  Finds the index of the maximum value in the flattened array.

4.2.1 Function Documentation

4.2.1.1 _apply_func_to_array()

```
mininumpy.array._apply_func_to_array (
            a,
            func)  [protected]
```

Helper to apply a function elementwise to an array.

Parameters

a	The input Array.
func	The function to apply.

Returns

A new Array with the function applied to each element.

4.2.1.2 _create_nested_list()

```
mininumpy.array._create_nested_list (
              shape,
              value)  [protected]
```

Helper to create a nested list of a given shape filled with a value.

Parameters

shape	The desired shape tuple.
value	The value to fill the list with.

Returns

A nested list.

4.2.1.3 abs()

```
mininumpy.array.abs (
              a)
```

Computes the absolute value of all elements.

4.2.1.4 arange()

```
mininumpy.array.arange (
              start,
              stop = None,
              step = 1)
```

Creates an array with a range of values.

Parameters

start	The start of the interval.
stop	The end of the interval.
step	The step size.

Returns

A 1D Array with the specified range of values.

4.2.1.5 argmax()

```
mininumpy.array.argmax (
            a)
```

Finds the index of the maximum value in the flattened array.

4.2.1.6 argmin()

```
mininumpy.array.argmin (
            a)
```

Finds the index of the minimum value in the flattened array.

4.2.1.7 array()

```
mininumpy.array.array (
            data)
```

Creates an Array object from a list or nested list.

Parameters

data	The input list.

Returns

An Array object.

4.2.1.8 exp()

```
mininumpy.array.exp (
            a)
```

Computes the exponential of all elements in the input array.

4.2.1.9 eye()

```
mininumpy.array.eye (
            n)
```

Creates an N x N identity matrix.

Parameters

n	The number of rows and columns.

Returns

An N x N identity Array.

4.2.1.10 linspace()

```
mininumpy.array.linspace (
            start,
            stop,
            num)
```

Creates an array with a specified number of evenly spaced values.

Parameters

start	The starting value of the sequence.
stop	The ending value of the sequence.
num	The number of samples to generate.

Returns

A 1D Array of evenly spaced samples.

4.2.1.11 log()

```
mininumpy.array.log (
              a)
```

Computes the natural logarithm of all elements in the input array.

4.2.1.12 max()

```
mininumpy.array.max (
              a)
```

Finds the maximum value among all array elements.

4.2.1.13 mean()

```
mininumpy.array.mean (
              a)
```

Computes the arithmetic mean of all array elements.

4.2.1.14 min()

```
mininumpy.array.min (
              a)
```

Finds the minimum value among all array elements.

4.2.1.15 ones()

```
mininumpy.array.ones (
              shape)
```

Creates an array of a given shape, filled with ones.

Parameters

shape	A tuple of ints specifying the shape.

Returns

An Array object filled with ones.

4.2.1.16 sqrt()

```
mininumpy.array.sqrt (
              a)
```

Computes the non-negative square-root of all elements.

4.2.1.17 sum()

```
mininumpy.array.sum (
              a)
```

Computes the sum of all array elements.

4.2.1.18 zeros()

```
mininumpy.array.zeros (
              shape)
```

Creates an array of a given shape, filled with zeros.

Parameters

shape	A tuple of ints specifying the shape.

Returns

An Array object filled with zeros.

4.3 mininumpy.linalg Namespace Reference

MiniNumPy Linear Algebra Module.

Functions

- dot (a, b)

  Computes the dot product of two arrays.
- matmul (a, b)

  Performs matrix multiplication on two 2D arrays.
- norm (a)

  Calculates the vector (L2) or matrix (Frobenius) norm.
- det (a)

  Calculates the determinant of a square 2D array.
- inv (a)

  Calculates the inverse of a square 2D array.
- eig (a)

  (Bonus) Placeholder for eigenvalue and eigenvector calculation.

4.3.1 Detailed Description

MiniNumPy Linear Algebra Module.

Contains functions for linear algebra operations like dot product, matrix multiplication, determinant, and inverse.

4.3.2 Function Documentation

4.3.2.1 det()

```
mininumpy.linalg.det (
            a)
```

Calculates the determinant of a square 2D array.

This implementation uses recursive Laplace expansion. It is not efficient for large matrices but is simple to implement.

Parameters

a	A square 2D Array.

Returns

The determinant of the array. @raises ValueError if the array is not square or not 2D.

4.3.2.2 dot()

```
mininumpy.linalg.dot (
            a,
            b)
```

Computes the dot product of two arrays.

- If both are 1D, it is the inner product.
- If both are 2D, it is matrix multiplication.

Parameters

a	First input array.
b	Second input array.

Returns

The dot product of a and b. @raises ValueError if dimensions are incompatible.

4.3.2.3 eig()

```
mininumpy.linalg.eig (
              a)
```

(Bonus) Placeholder for eigenvalue and eigenvector calculation.

Calculating eigenvalues is a complex numerical problem (e.g., using QR algorithm) and is beyond the scope of a simple implementation from scratch.

Parameters

a	A square 2D Array. @raises NotImplementedError

4.3.2.4 inv()

```
mininumpy.linalg.inv (
              a)
```

Calculates the inverse of a square 2D array.

The inverse is calculated using the formula: inv(A) = (1/det(A)) $*$ adj(A), where adj(A) is the adjugate matrix (transpose of the cofactor matrix).

Parameters

a	A square, invertible 2D Array.

Returns

The inverse of the array. @raises ValueError if the array is not square or is singular (det=0).

4.3.2.5 matmul()

```
mininumpy.linalg.matmul (
              a,
              b)
```

Performs matrix multiplication on two 2D arrays.

Parameters

a	A 2D Array with shape (m, n).
b	A 2D Array with shape (n, p).

Returns

A new 2D Array with shape (m, p). @raises ValueError if arrays are not 2D or shapes are incompatible.

4.3.2.6 norm()

```
mininumpy.linalg.norm (
              a)
```

Calculates the vector (L2) or matrix (Frobenius) norm.

Parameters

a	The input array.

Returns

The norm of the array.

Chapter 5

Class Documentation

5.1 mininumpy.array.Array Class Reference

A simplified multi-dimensional array class, similar to NumPy's ndarray.

Public Member Functions

- __init__ (self, data)

  Constructor for the Array class.
- __str__ (self)

  Provides a user-friendly string representation of the array.
- __repr__ (self)

  Provides an unambiguous string representation of the array object.
- reshape (self, new_shape)

  Returns a new Array with a different shape, without changing its data.
- transpose (self)

  Transposes the array.
- __add__ (self, other)
- __sub__ (self, other)
- __mul__ (self, other)
- __truediv__ (self, other)
- __pow__ (self, other)
- __matmul__ (self, other)

  Implements the matrix multiplication operator (@).
- __rsub__ (self, other)

  Implements the right-side subtraction (e.g., scalar - array).

Public Attributes

- data = data
- shape = self._derive_shape(data)
- ndim = len(self.shape)
- size = reduce(operator.mul, self.shape, 1)

Protected Member Functions

- _derive_shape (self, data)

  Recursively determines the shape of the nested list data.
- _validate_shape (self, data, shape, level)

  Recursively validates that all sub-arrays at a certain depth have the same length.
- _flatten (self)

  Flattens the multi-dimensional data into a 1D list.
- _elementwise_op (self, other, op)

  A helper function to perform elementwise operations.

5.1.1 Detailed Description

A simplified multi-dimensional array class, similar to NumPy's ndarray.

This class wraps a Python list (or nested lists) and provides methods for basic array manipulation, such as shape, transpose, and elementwise operations.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 __init__()

```
mininumpy.array.Array.__init__ (
            self,
            data)
```

Constructor for the Array class.

Initializes an array from a list or nested list. It validates the data, determines the shape, and calculates the total size.

Parameters

data	A Python list or nested list. @raises TypeError if data is not a list. @raises ValueError if rows have inconsistent lengths.

5.1.3 Member Function Documentation

5.1.3.1 __add__()

```
mininumpy.array.Array.__add__ (
            self,
            other)
```

5.1.3.2 __matmul__()

```
mininumpy.array.Array.__matmul__ (
            self,
            other)
```

Implements the matrix multiplication operator (@).

Parameters

other	The other Array to multiply with.

Returns

The result of the matrix multiplication.

5.1.3.3 __mul__()

```
mininumpy.array.Array.__mul__ (
            self,
            other)
```

5.1.3.4 __pow__()

```
mininumpy.array.Array.__pow__ (
            self,
            other)
```

5.1.3.5 __repr__()

```
mininumpy.array.Array.__repr__ (
            self)
```

Provides an unambiguous string representation of the array object.

Returns

A string representation of the object.

5.1.3.6 __rsub__()

```
mininumpy.array.Array.__rsub__ (
            self,
            other)
```

Implements the right-side subtraction (e.g., scalar - array).

Parameters

other	The other operand (scalar).

Returns

The result of the elementwise subtraction.

5.1.3.7 __str__()

```
mininumpy.array.Array.__str__ (
            self)
```

Provides a user-friendly string representation of the array.

Returns

A string that pretty-prints the array.

5.1.3.8 __sub__()

```
mininumpy.array.Array.__sub__ (
            self,
            other)
```

5.1.3.9 __truediv__()

```
mininumpy.array.Array.__truediv__ (
            self,
            other)
```

5.1.3.10 _derive_shape()

```
mininumpy.array.Array._derive_shape (
            self,
            data)  [protected]
```

Recursively determines the shape of the nested list data.

Parameters

data	The list or nested list.

Returns

A tuple representing the shape.

5.1.3.11 _elementwise_op()

```
mininumpy.array.Array._elementwise_op (
            self,
            other,
            op)  [protected]
```

A helper function to perform elementwise operations.

Handles operations with a scalar or another array of the same shape.

Parameters

other	The other operand (scalar or Array).
op	A function (e.g., from the operator module) to apply.

Returns

A new Array with the result of the operation. @raises ValueError if arrays have mismatched shapes.

5.1.3.12 _flatten()

```
mininumpy.array.Array._flatten (
              self)  [protected]
```

Flattens the multi-dimensional data into a 1D list.

Returns

A 1D list containing all elements of the array.

5.1.3.13 _validate_shape()

```
mininumpy.array.Array._validate_shape (
              self,
              data,
              shape,
              level)  [protected]
```

Recursively validates that all sub-arrays at a certain depth have the same length.

Parameters

data	The data to validate.
shape	The target shape.
level	The current recursion depth. @raises ValueError if shapes are inconsistent.

5.1.3.14 reshape()

```
mininumpy.array.Array.reshape (
              self,
              new_shape)
```

Returns a new Array with a different shape, without changing its data.

The new shape must be compatible with the original size.

Parameters

new_shape	A tuple of ints giving the new shape.

Returns

A new Array object with the specified shape. @raises ValueError if the new shape is not compatible with the array's size.

5.1.3.15 transpose()

```
mininumpy.array.Array.transpose (
              self)
```

Transposes the array.

Currently only supports 2D arrays.

Returns

A new Array object that is the transpose of the original. @raises ValueError if the array is not 2D.

5.1.4 Member Data Documentation

5.1.4.1 data

```
mininumpy.array.Array.data = data
```

5.1.4.2 ndim

```
mininumpy.array.Array.ndim = len(self.shape)
```

5.1.4.3 shape

```
mininumpy.array.Array.shape = self._derive_shape(data)
```

5.1.4.4 size

```
mininumpy.array.Array.size = reduce(operator.mul, self.shape, 1)
```

The documentation for this class was generated from the following file:

- mininumpy/array.py

Chapter 6

File Documentation

6.1 mininumpy/__init__.py File Reference

Namespaces

- namespace mininumpy

  MiniNumPy package initializer.

6.2 mininumpy/array.py File Reference

Classes

- class mininumpy.array.Array

  A simplified multi-dimensional array class, similar to NumPy's ndarray.

Namespaces

- namespace mininumpy

  MiniNumPy package initializer.
- namespace mininumpy.array

Functions

- mininumpy.array.array (data)

  Creates an Array object from a list or nested list.
- mininumpy.array._create_nested_list (shape, value)

  Helper to create a nested list of a given shape filled with a value.
- mininumpy.array.zeros (shape)

  Creates an array of a given shape, filled with zeros.
- mininumpy.array.ones (shape)

  Creates an array of a given shape, filled with ones.
- mininumpy.array.eye (n)

  Creates an N x N identity matrix.

- mininumpy.array.arange (start, stop=None, step=1)

  Creates an array with a range of values.
- mininumpy.array.linspace (start, stop, num)

  Creates an array with a specified number of evenly spaced values.
- mininumpy.array._apply_func_to_array (a, func)

  Helper to apply a function elementwise to an array.
- mininumpy.array.exp (a)

  Computes the exponential of all elements in the input array.
- mininumpy.array.log (a)

  Computes the natural logarithm of all elements in the input array.
- mininumpy.array.sqrt (a)

  Computes the non-negative square-root of all elements.
- mininumpy.array.abs (a)

  Computes the absolute value of all elements.
- mininumpy.array.sum (a)

  Computes the sum of all array elements.
- mininumpy.array.mean (a)

  Computes the arithmetic mean of all array elements.
- mininumpy.array.min (a)

  Finds the minimum value among all array elements.
- mininumpy.array.max (a)

  Finds the maximum value among all array elements.
- mininumpy.array.argmin (a)

  Finds the index of the minimum value in the flattened array.
- mininumpy.array.argmax (a)

  Finds the index of the maximum value in the flattened array.

6.3 mininumpy/linalg.py File Reference

Namespaces

- namespace mininumpy

  MiniNumPy package initializer.
- namespace mininumpy.linalg

  MiniNumPy Linear Algebra Module.

Functions

- mininumpy.linalg.dot (a, b)

  Computes the dot product of two arrays.
- mininumpy.linalg.matmul (a, b)

  Performs matrix multiplication on two 2D arrays.
- mininumpy.linalg.norm (a)

  Calculates the vector (L2) or matrix (Frobenius) norm.
- mininumpy.linalg.det (a)

  Calculates the determinant of a square 2D array.
- mininumpy.linalg.inv (a)

  Calculates the inverse of a square 2D array.
- mininumpy.linalg.eig (a)

  (Bonus) Placeholder for eigenvalue and eigenvector calculation.

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